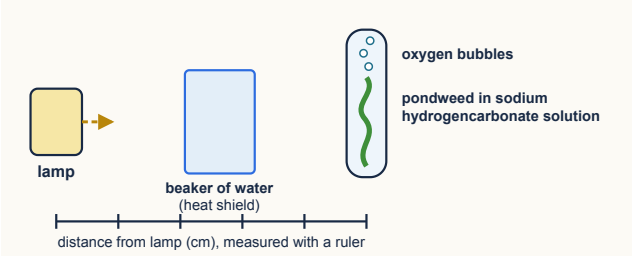
Pondweed practical: does more light mean more photosynthesis?

Name _____ Date _____

Aim: to find out how light intensity affects the rate of photosynthesis in pondweed.

> **Safety:** the lamp gets hot, so don't touch the bulb. Keep water away from the lamp and plug. Dry wet hands before touching anything electrical. Wipe up spills.



Progress check 1 · before you switch on. Fill in: C (I change) _____ M1 (I measure) _____ One S (I keep the same) _____ by _____

Method

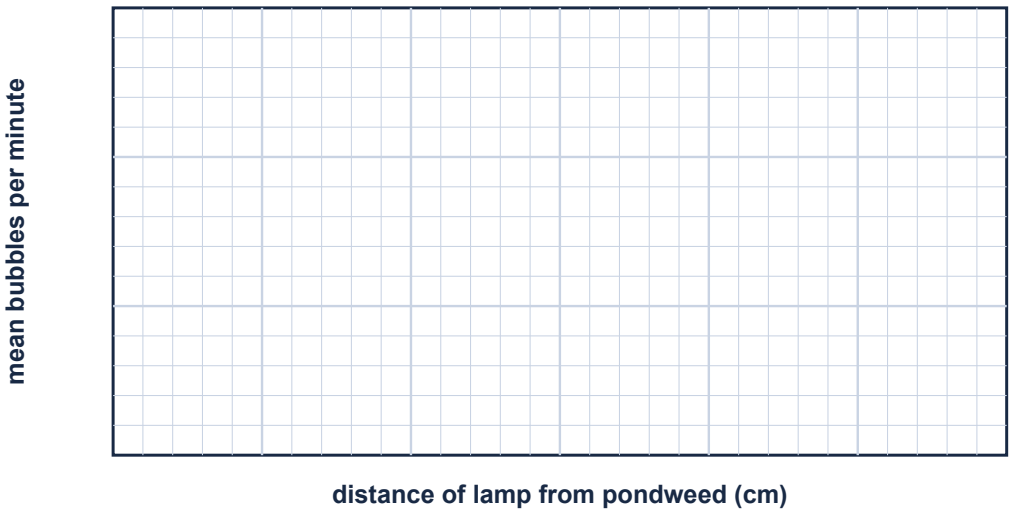
1. Fill a boiling tube with the sodium hydrogencarbonate solution provided. It supplies carbon dioxide.
2. Cut a 5 cm piece of pondweed. Put it in the tube with the **cut end at the top**.
3. Stand the tube in the rack. Put the beaker of water between the lamp and the tube. Turn the room lights off.
4. Use the ruler to place the lamp **10 cm** from the pondweed. Record the water temperature.
5. Wait **2 minutes** for the pondweed to adjust. Then count the bubbles for **1 minute**.
6. Repeat the 1-minute count twice more at this distance.
7. Move the lamp to 20, 30, 40 and 50 cm. Repeat steps 4–6 at each distance.

Results

Distance (cm)	Count 1	Count 2	Count 3	Mean bubbles per minute	Temp (°C)
10					
20					
30					
40					
50					

Progress check 2 · after 10 cm. Show your mean working for 10 cm. Circle any count very different from the others. What will you do about it?

Plot your results



1. Describe the pattern in your results. Use numbers from your table. **[2]**

2. Explain the pattern. Use the words *light energy*, *chlorophyll* and *oxygen*. **[3]**

3. Why does counting oxygen bubbles show the rate of photosynthesis? **[1]**

4. Bubbles can be different sizes. Explain why this is a problem, and suggest a better way to measure the oxygen. **[2]**

5. **Think deeper.** Very close to the lamp, the rate may stop rising. Suggest a reason. **[1]**

Progress check 3 · on your own, no notes. Another group had no beaker of water. Their water warmed from 22 °C to 30 °C when the lamp was at 10 cm. They concluded: “More light

made the pondweed photosynthesise faster." Explain why their conclusion might not be valid. **[2]**